PSEUDO-COHERENCE IDENTIFIES WITH COHERENT COHOMOLOGY ON NOETHERIAN SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- On an affine scheme D_QCoh is the derived category of modules and the functor identifies cohomology. invoked at proof lines 2496-2497. Represents E by a complex of A-modules M.
- On affines m-pseudo-coherence of an object agrees with m-pseudo-coherence of its module complex. invoked at proof lines 2498-2499. Transfers condition (2) to M.
- A coherent sheaf on an affine Noetherian scheme is a finite module and conversely. invoked at proof lines 2500-2502. Transfers condition (1) to finite modules.
- For a complex of modules over a Noetherian ring m-pseudo-coherence is equivalent to coherent cohomology in degrees at least m and vanishing above. invoked at proof lines 2503-2504. Finishes the equivalence of (1) and (2).

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 2496-2497 to represent E by a complex of A-modules; the transport and conclusion are inside the counted span.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Cited at proof lines 2498-2499 to transfer condition (2); the equivalence is assembled inside the counted span.

Source reference. Original lemma 01PB: sources/stacks-chow/tags.txt, source lines 2224–2224. Cited at proof lines 2500-2502 to transfer condition (1); the equivalence is assembled inside the counted span.

Source reference. Original lemma 066E: sources/stacks-chow/tags.txt, source lines 7933–7933. Cited at proof lines 2503-2504 as the final module-level input; the assembly of both directions is inside the counted span.

Lemma 0.2. *Let X be a Noetherian scheme. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. For $m \in \mathbf{Z}$ the following are equivalent*

(1) *$H^i(E)$ is coherent for $i \geq m$ and zero for $i \gg 0$, and*

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(2) *E is m-pseudo-coherent.*

In particular, E is pseudo-coherent if and only if E is an object of $D^-_{Coh}(\mathcal{O}_X)$.

Proof. As X is quasi-compact we see that in both (1) and (2) the object E is bounded above. Thus the question is local on X and we may assume X is affine. Say $X = \mathrm{Spec}(A)$ for some Noetherian ring A. In this case E corresponds to a complex of A-modules $M^\bullet$ by Lemma 06Z0. By Lemma 08E7 we see that E is m-pseudo-coherent if and only if $M^\bullet$ is m-pseudo-coherent. On the other hand, $H^i(E)$ is coherent if and only if $H^i(M^\bullet)$ is a finite A-module (Properties, Lemma 01PB). Thus the result follows from More on Algebra, Lemma 066E. □